

```
alysa@ubuntu-ker-vm:~$ cd /home/alysa/
alysa@ubuntu-ker-vm:~$ wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.6.10.tar.xz
--2025-10-06 00:10:59-- https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.6.10.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 199.232.173.176, 2a04:4e42:6b::432
Connecting to cdn.kernel.org (cdn.kernel.org)|199.232.173.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 140042308 (134M) [application/x-xz]
Saving to: 'linux-6.6.10.tar.xz'

linux-6.6.10.tar.xz      100%[=====>]
2025-10-06 00:11:25 (5.21 MB/s) - 'linux-6.6.10.tar.xz' saved [140042308/140042308]

alysa@ubuntu-ker-vm:~$
```

tar -xf linux-6.6.10.tar.xz

```
alyssa@ubuntu-ker-vm:~$ tar -xf linux-6.6.10.tar.xz
alyssa@ubuntu-ker-vm:~$
```

заходим в директорию
make tinyconfig

```
alyssa@ubuntu-ker-vm:~$ tar -xf linux-6.6.10.tar.xz
alyssa@ubuntu-ker-vm:~$ cd linux-6.6.10/
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ make tinyconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
#
# configuration written to .config
#
Using .config as base
```

make menuconfig , а то после tiny ничего не соберется
а еще

```
x86_64
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371SB PIIX3 IDE [Natoma/Triton II]
00:01.3 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 03)
00:02.0 VGA compatible controller: Device 1234:1111 (rev 02)
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 03)
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ lsusb
```

e1000 драйвер включаем

```
[*] Intel (82500/82503/82506) devices (NEW)
[*] Intel devices (NEW)
< > Intel(R) PRO/100+ support (NEW)
[*] Intel(R) PRO/1000 Gigabit Ethernet support
< > Intel(R) PRO/1000 PCI-Express Gigabit Ethernet support (NEW)
< > Intel(R) 82575/82576 PCI-Express Gigabit Ethernet support (NEW)
< > Intel(R) 82576 Virtual Function Ethernet support (NEW)
```

--- Block devices

```
< > Null test block driver
< > Block Device Driver for Micron PCIe SSDs
< > Loopback device support
      *** DRBD disabled because PROC_FS or INET not selected ***
< > Network block device support
<*> RAM block device support
(16) Default number of RAM disks (NEW)
(4096) Default RAM disk size (kbytes) (NEW)
< > Packet writing on CD/DVD media (DEPRECATED)
< > ATA over Ethernet support
<*> Virtio block driver
< > Userspace block driver (Experimental)
```

включила базовые опции

[] Compile also drivers which will not load

```
[ ] Compile the kernel with warnings as errors
() Local version - append to kernel release
[ ] Automatically append version information to the version string
() Build ID Salt
      Kernel compression mode (XZ) --->
() Default init path
((none)) Default hostname
[*] System V IPC
[ ] General notification queue
[ ] Enable process_vm_readv/writev syscalls
[*] uselib syscall (for libc5 and earlier)
      IRQ subsystem ----
      Timers subsystem --->
      BPF subsystem --->
      Preemption Model (No Forced Preemption (Server)) --->
[ ] Preemption behaviour defined on boot
      CPU/Task time and stats accounting --->
      RCU Subsystem --->
<*> Kernel .config support
(17) Kernel log buffer size (16 => 64KB, 17 => 128KB) (NEW)
      Scheduler features ----
[*] Control Group support --->
[*] Namespaces support --->
[ ] Automatic process group scheduling
[ ] Kernel->user space relay support (formerly relayfs)
[*] Initial RAM filesystem and RAM disk (initramfs/initrd) support
() Initramfs source file(s) (NEW)
[*] Support initial ramdisk/ramfs compressed using gzip (NEW)
[*] Support initial ramdisk/ramfs compressed using bzip2 (NEW)
[*] Support initial ramdisk/ramfs compressed using LZMA (NEW)
[*] Support initial ramdisk/ramfs compressed using XZ (NEW)
[*] Support initial ramdisk/ramfs compressed using LZ0 (NEW)
[*] Support initial ramdisk/ramfs compressed using LZ4 (NEW)
```

```

[*] Support initial ramdisk/ramfs compressed using LZ4 (NEW)
[*] Support initial ramdisk/ramfs compressed using ZSTD (NEW)
[ ] Boot config support
[ ] Preserve cpio archive mtimes in initramfs
Compiler optimization level (Optimize for size (-Os)) --->
[*] Configure standard kernel features (expert users) --->
[ ] Enable kcmp() system call
[ ] Enable rseq() system call
[ ] Enable cachestat() system call
[ ] PC/104 support
Kernel Performance Events And Counters --->
[ ] Profiling support
Kexec and crash features --->

```

```

[ ] Preserve cpio archive mtimes in initramfs
Compiler optimization level (Optimize for size (-Os)) --->
[*] Configure standard kernel features (expert users) --->
[ ] Enable kcmp() system call

```

```

--- Configure standard kernel features (expert users)
[*] Multiple users, groups and capabilities support
[*] sgetmask/ssetmask syscalls support
[*] Sysfs syscall support
[*] open by fhandle syscalls
[*] Posix Clocks & timers
[*] Enable support for printk
[*] BUG() support
[ ] Enable PC-Speaker support
[*] Enable full-sized data structures for core
[*] Enable futex support
[*] Enable eventpoll support
[*] Enable signalfd() system call
[*] Enable timerfd() system call
[*] Enable eventfd() system call
[*] Use full shmem filesystem
[*] Enable AIO support
[*] Enable IO uring support
[*] Enable madvise/fadvise syscalls
[*] Enable membarrier() system call
[ ] Load all symbols for debugging/ksymoops

```

```

    Timer tick handling (Periodic timer ticks (constant rate, no dynticks)) --->
[*] Old Idle dynticks config
[ ] High Resolution Timer Support
(125) Clocksource watchdog maximum allowable skew (in s)

```

```

[*] Suspend to RAM and standby
[ ] Skip kernel's sys_sync() on suspend to RAM/standby (NEW)
[ ] Opportunistic sleep (NEW)
[ ] Userspace opportunistic sleep (NEW)
[ ] User space wakeup sources interface (NEW)
-* Device power management core functionality
[ ] Power Management Debug Support (NEW)
[ ] Enable workqueue power-efficient mode by default (NEW)
[*] Energy Model for devices with DVFS (CPUs, GPUs, etc)
[*] ACPI (Advanced Configuration and Power Interface) Support --->
[*] Power Management Timer Support (NEW)
    CPU Frequency scaling --->
    CPU Idle --->
[*] Cpuidle Driver for Intel Processors

```

```

-* CPU Frequency scaling
[ ] CPU frequency transition statistics (NEW)
    Default CPUFreq governor (schedutil) --->
-* 'performance' governor
<> 'powersave' governor (NEW)
<> 'userspace' governor for userspace frequency scaling (NEW)
<> 'ondemand' cpufreq policy governor (NEW)
<> 'conservative' cpufreq governor (NEW)
-* 'schedutil' cpufreq policy governor
    *** CPU frequency scaling drivers ***
-* Intel P state control
<> Processor Clocking Control interface driver (NEW)
[*] AMD Processor P-State driver
(3) AMD Processor P-State default mode (NEW)
<> selftest for AMD Processor P-State driver (NEW)
<> ACPI Processor P-States driver (NEW)
<> Intel Enhanced SpeedStep (deprecated) (NEW)
<> Intel Pentium 4 clock modulation (NEW)
    *** shared options ***

```

```

[ ] ISA bus support on modern systems
[ ] ISA-style DMA support

```

Bus options отключаем

```

[ ] Kprobes (NEW)
[*] Optimize very unlikely/likely branches
[ ] Static key selftest (NEW)
[ ] Static call selftest
[ ] Enable seccomp to safely execute untrusted bytecode
[*] Stack Protector buffer overflow detection
[*] Strong Stack Protector (NEW)
    Link Time Optimization (LTO) (None) --->
(28) Number of bits to use for ASLR of mmap base address
[ ] Provide system calls for 32-bit time_t
[*] Use a virtually-mapped stack (NEW)
[ ] Support for randomizing kernel stack offset on syscall entry
_ GCov-based kernel profiling ----

```

```

Executable file formats
the menu. <Enter> selects submenus ---> (or empty submenus -
Y> includes, <N> excludes, <M> modularizes features. Press <
[*] built-in [ ] excluded <M> module < > module capable

[*] Kernel support for ELF binaries
<*> Kernel support for scripts starting with #!
< > Kernel support for MISC binaries
[ ] Enable core dump support

```

```

[*] Support for paging of anonymous memory (swap) (NEW) --->
    SLAB allocator options --->
[ ] Page allocator randomization
[ ] Disable heap randomization
-* Sparse Memory virtual memmap
[ ] Memory hotplug (NEW) ----
[ ] Allow for memory compaction
[*] Free page reporting
[ ] Enable KSM for page merging
(4096) Low address space to protect from user allocation
[ ] Transparent Hugepage Support ----
[ ] Contiguous Memory Allocator
[ ] Defer initialisation of struct pages to kthreads (NEW)
[ ] Support DMA zone
-* Support DMA32 zone
[*] Enable VM event counters for /proc/vmstat
[ ] Collect percpu memory statistics
    *** GUP_TEST needs to have DEBUG_FS enabled ***
<*> Enable a module to run time tests on dma_pool
[ ] Enable memfd_create() system call
[ ] Enable memfd_secret() system call
[ ] Enable userfaultfd() system call
[ ] Multi-Gen LRU
    Data Access Monitoring --->

```

```

[*] Support for uevent helper
()   path to uevent helper (NEW)
[*] Maintain a devtmpfs filesystem to mount at /dev
[ ]   Automount devtmpfs at /dev, after the kernel mounted the rootfs (NEW)
[ ]   Use nosuid,noexec mount options on devtmpfs (NEW)
[ ]   Select only drivers that don't need compile-time external firmware
[ ]   Disable drivers features which enable custom firmware building
      Firmware loader --->
[ ]   Allow device coredump
[ ]   Driver Core verbose debug messages
[ ]   Managed device resources verbose debug messages
[ ]   Test driver remove calls during probe (UNSTABLE)
< > Build kernel module to test asynchronous driver probing (NEW)
[*] sync_state() behavior defaults to timeout instead of strict

```

```

      ARM System Control and Management Interface Protocol ----
< > BIOS Enhanced Disk Drive calls determine boot disk
[*] Add firmware-provided memory map to sysfs
[*] Mark VGA/VBE/EFI FB as generic system framebuffer
[ ] Google Firmware Drivers ----
_ Tegra firmware driver ----

```

```

[ ] Enable access key retention support
[ ] Restrict unprivileged access to the kernel syslog
[ ] Enable the securityfs filesystem
[*] Harden memory copies between kernel and userspace
[*] Harden common str/mem functions against buffer overflows
[ ] Force all usermode helper calls through a single binary
      First legacy 'major LSM' to be initialized (Unix Discretionary Access Controls) --->
(landlock,lockdown,yama,loadpin,safesetid,bpf) Ordered list of enabled LSMs
      Kernel hardening options --->

```

make -j\$(nproc)

make modules -j\$(nproc)

```

alyssa@ubuntu-ker-vm:~/linux-6.6.10$ make modules -j$(nproc)
CALL      scripts/checksyscalls.sh
DESCEND   objtool
INSTALL   libsubcmd_headers
alyssa@ubuntu-ker-vm:~/linux-6.6.10$

```

sudo make modules_install

```
6.6.0-85-generic
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ sudo make modules_install
SYMLINK /lib/modules/6.6.10/build
INSTALL /lib/modules/6.6.10/modules.order
INSTALL /lib/modules/6.6.10/modules.builtin
INSTALL /lib/modules/6.6.10/modules.builtin.modinfo
INSTALL /lib/modules/6.6.10/kernel/drivers/thermal/intel/x86_pkg_temp_thermal.ko
DEPMOD /lib/modules/6.6.10
alyssa@ubuntu-ker-vm:~/linux-6.6.10$
```

//на рил системе убедиться до make install, что есть загрузчик в boot (дир grub)

sudo make install

```
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ sudo make install
INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.6.10 /boot/vmlinuz-6.6.10
update-initramfs: Generating /boot/initrd.img-6.6.10
[12569.474639] I/O error, dev fd0, sector 0 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 6.6.10 /boot/vmlinuz-6.6.10
run-parts: executing /etc/kernel/postinst.d/update-notifier 6.6.10 /boot/vmlinuz-6.6.10
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 6.6.10 /boot/vmlinuz-6.6.10
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 6.6.10 /boot/vmlinuz-6.6.10
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-85-generic
Found initrd image: /boot/initrd.img-6.8.0-85-generic
Found linux image: /boot/vmlinuz-6.6.10
Found initrd image: /boot/initrd.img-6.6.10
Found linux image: /boot/vmlinuz-6.6.10.old
Found initrd image: /boot/initrd.img-6.6.10
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ _
```

Ядро собралось!!!

```
Adding boot menu entry for UEFI Firmware Settings ...
done
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ sudo nano /etc/default/grub_
```




```

GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=menu
GRUB_TIMEOUT=5
GRUB_DISTRIBUTOR=`( . /etc/os-release; echo ${NAME:-Ubuntu} ) 2>/dev/null ||
GRUB_CMDLINE_LINUX_DEFAULT=""
GRUB_CMDLINE_LINUX=""

# If your computer has multiple operating systems installed, then you
# probably want to run os-prober. However, if your computer is a host
# for guest OSes installed via LVM or raw disk devices, running
# os-prober can cause damage to those guest OSes as it mounts
# filesystems to look for things.
#GRUB_DISABLE_OS_PROBER=false

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo'
#GRUB_GFXMODE=640x480

# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"

alyssa@ubuntu-ker-vm:~$ sudo update-grub

```

sudo update-grub

```

done
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-85-generic
Found initrd image: /boot/initrd.img-6.8.0-85-generic
Found linux image: /boot/vmlinuz-6.6.10
Found initrd image: /boot/initrd.img-6.6.10
Found linux image: /boot/vmlinuz-6.6.10.old
Found initrd image: /boot/initrd.img-6.6.10
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
alyssa@ubuntu-ker-vm:~/linux-6.6.10$ _

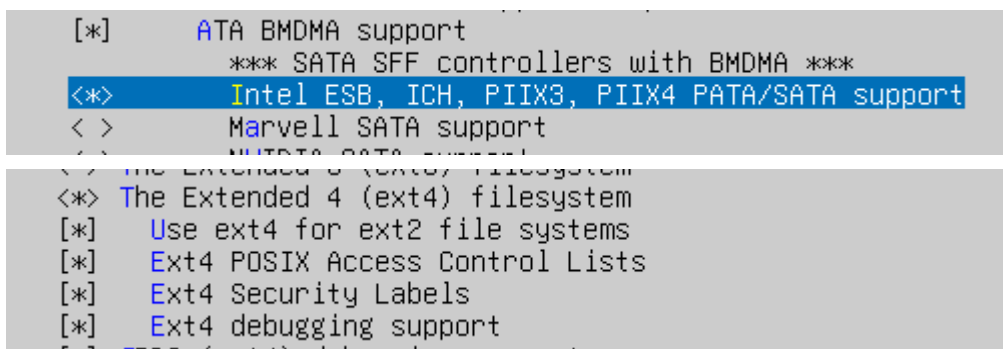
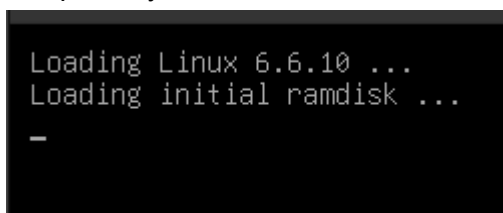
```

sudo reboot



6.6.10 - новое

застряли тут:



заново

\$(nproc) у меня ==1, so делаю без флага -j



не вышло запуститься